

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Shorter / different 1 ^o structure/ less amino acids/ smaller/ ORA (1)	1			1		
		(iii)		do not have to repeat treatment/ more permanent treatment/ change present in daughter cells/ change present after cell replicates/ can pass to next generation (1)	1			1		
		(iv)		Unknown long-term effects /possible activation of oncogenes / modified gene passed on to next generation/ affect {other genes/ later generations}(1)	1			1		
	(e)			Not sex linked / on autosomes (1) Dominant/not recessive (1) Three examples to justify the conclusions for (3) If recessive 1 + 2 could not produce 6/ unaffected {female/ child} 8 + 9 could not produce 13 /14/ unaffected {female/ child} OR must be dominant, otherwise all children of 1+ 2 / 8 + 9 would be affected If sex linked 1 + 2 could not produce 6/ unaffected female 8 + 9 could not produce 13 or 14/ unaffected female 4 + 5 could not produce 10/ an affected male OR must be autosomal, otherwise all daughters of 1+ 2 / 8 + 9 would be affected Max (4) if examples are given to only support one conclusion.			5	5		
				Question 2 total	8	7	5	20		